Finite Element Analysis of a Plate under Cyclic Axial Load

1 introduction

Cyclic loading is common in automotive, aerospace, and civil structures and often governs fatigue life. Finite Element Analysis (FEA) offers a flexible approach to quantifying stress-strain histories. This report consolidates the theoretical derivations, code implementation, verification studies, and 2D static results into a coherent narrative.

Project objectives

- Derive FEM governing equations for 1D bar and 2D triangular elements
- Implement a Python-based solver for a static and dynamic analysis
- Validates against analytical solutions and ANSYS
- Extend to 2D transient simulation and assess mesh sensitivity.

2 Theoretical Background

2.1 Weight Residual Method: Galerkin Approach

Based on the Weight Residual Method (WRM), the mathematical equation can be rewritten as

$$\int_0^L W_i R \, dx = 0 \tag{1}$$

Which residual R is equal to the difference between the equation and the mathematical model

$$R(x) = \frac{d}{dx}\left(EA\frac{du}{dx} + f(x)\right)$$

$$\int_0^L W_i \cdot \frac{d}{dx}\left(EA\frac{du}{dx} + f(x)\right) dx = 0$$

$$\int_0^L W_i \frac{d}{dx}\left(EA\frac{du}{dc}\right) dx + \int_0^L W_i f(x) dx = 0$$

By applying integration by parts on the first part of the equation

$$\int_0^L W_i \frac{du}{dx}\left(EA\frac{du}{dx}\right) dx = W_i EA\frac{du}{dx}\Big|_0^L - \int_0^L EA\frac{du}{dx}\frac{dW_i}{dx} dx \tag{2}$$

Then by substituting it into the above equation,

$$W_i EA\frac{du}{dx}\Big|_0^L - \int_0^L EA\frac{du}{dx}\frac{dW_i}{dx} dx + \int_0^L W_i f(x) dx = 0$$

$$\int_0^L EA\frac{dW_i}{dx}\frac{du}{dx} dx = W_i EA\frac{du}{dx}\Big|_0^L + \int_0^L W_i f(x) dx \tag{3}$$

By using Galerkin application ($W_i = N_i$)

$$\int_0^L EA\frac{dN_i}{dx}\frac{du}{dx} dx = N_i EA\frac{du}{dx}\Big|_0^L + \int_0^L N_i f(x) dx = F_{boundery} + F_f \tag{4}$$

2.2 Governing equations: Axial loading case

2.2.1 1D Mathematical Model

1. *Strong form (static and dynamic)*

$$EA\frac{d^2u}{dx^2} + f(x) = \rho A\ddot{u}(x,t) \tag{5}$$

where

- $u(x,t)$ displacement; E young's modulus; A cross-sectional area
- f body force per unit length; ρ material density
- Dots denote time derivation (dynamic case); for static analysis, the right-hand inertia term is zero.

2. *Residual and weighted residual statement*

Define the residual

$$R(u) = EAu'' + f - \rho A\ddot{u}$$

Galerkin chooses the weighting function equal to the shape and test function $N_i(x)$:

$$\int_0^L N_i R(u)\, dx = 0, \quad i = 1, \dots, n_{nodes} \tag{6}$$

3. *Integration by part (reduce derivation order)*

Only second-derivation term needs reduction:

$$\int_0^L N_i EAu''\, dx = [N_i EAu']_0^L - \int_0^L EAu' N_i'\, dx \tag{7}$$

4. *Boundary conditions emerge*

- Prescribe Boundary Condition (PBC) at the fixed end $x = 0$:

$$u(0) = 0 \Longrightarrow N_i(0) = 0$$

- Natural Boundary Condition (NBC) at $x = L$:

$$\text{axial traction } EAu'(L) = P(t)$$

thus,

$$[N_i EA\, u']_0^L = N_i(L)P(t)$$

5. *Final weak form*

$$\int_0^L EAu' N_i'\, dx = \int_0^L \rho A\ddot{u} N_i\, dx + \int_0^L f N_i\, dx + N_i(L)P(t) \tag{8}$$

6. *Discretization with linear 2-node element*

On an element of length L_e,

$$u(x) \approx N_1 u_1 + N_2 u_2$$

$$N_1 = \frac{x_2 - x}{L_e}, \quad N_2 = \frac{x - x_1}{L_e}$$

Element matrices:

$$K_e = \frac{EL}{L_e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad M_e^{consistence} = \frac{\rho A L_e}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad F_e = \int_{x_1}^{x_2} N^T f \, dx$$

Assembly over all elements yields equation

$$M\ddot{U} + C\dot{U} + KU = F(t) \tag{9}$$

where:

- M is the global mass matrix.
- C is the global damping matrix ($C = \alpha M + \beta K$ for Rayleigh damping).
- K is the global stiffness matrix.
- U is the global displacement matrix.
- $\dot{U}$ is the global displacement vector.
- $\ddot{U}$ is the acceleration vector.
- $F(t)$ is the time-dependent external vector ($F(t) = F_{body} + F_{traction}(t)$).

2.2.2 2D Mathematical Model

1. *Strong form (static and dynamic)*

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + f_x = \rho \frac{\partial^2 u_x}{\partial t^2} \quad \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + f_y = \rho \frac{\partial^2 u_y}{\partial t^2}$$

In vector form

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = \rho \ddot{\mathbf{u}}, \quad f = \begin{bmatrix} f_x \\ f_y \end{bmatrix}, \quad \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix}, \quad \boldsymbol{u} = \begin{bmatrix} u_x \\ u_y \end{bmatrix} \tag{10}$$

where:

- $\sigma_{xx}, \sigma_{yy}, \sigma_{xy}$ are the stress components.
- $u_x u_y$ are the displacement components in the x and y directions.
- f_x, f_y are the body force per unit area.
- ρ is the material density.

2. *Residual and Galerkin statement*

For each scalar shape function N_i:

$$\int_\Omega N_i(\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{f} - \rho\ddot{\boldsymbol{u}})\, d\Omega = 0 \tag{11}$$

3. *Divergence theorem (integration by part)*

$$-\int_\Omega \nabla N_i \cdot \boldsymbol{\sigma}\, d\Omega + \int_{\Gamma_t} N_i \boldsymbol{t}\, d\Gamma = \int_\Omega N_i \rho\ddot{\boldsymbol{u}}\, d\Omega + \int_\Omega N_i \boldsymbol{f}\, d\Omega \tag{12}$$

where $\boldsymbol{t} = \boldsymbol{\sigma n}$ is the prescribed traction on the NBC Γ_t

4. *Constitutive and kinematic relations*

$$\boldsymbol{\sigma} = D\boldsymbol{\epsilon} \quad \boldsymbol{\epsilon} = B\boldsymbol{u}$$

with the strain and plane-stress matric

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial u_x}{\partial x} \\ \dfrac{\partial u_y}{\partial_y} \\ \dfrac{\partial u_x}{\partial y} + \dfrac{\partial u_y}{\partial_y} \end{bmatrix} \quad D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix}$$

Substituting into equation (12)

$$\int_\Omega B^T DB\, \boldsymbol{u}\, d\Omega = \int_\Omega N^T \rho\ddot{\boldsymbol{u}}\, d\Omega + \int_\Omega N^T \boldsymbol{f}\, d\Omega + \int_{\Gamma_t} N^T \boldsymbol{t}\, d\Gamma \tag{13}$$

5. *Constan-stain triangle (CST) element*

$$B = \frac{1}{2A_e}\begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}, \quad K_e = tA_e B^T DB, \quad M_e^{consistent} = \rho t A_e \frac{1}{12}\begin{bmatrix} 2I_2 & I_2 & I_2 \\ I_2 & 2I_2 & I_2 \\ I_2 & I_2 & 2I_2 \end{bmatrix}$$

with $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. A lumped mass $M_e^{lumped} = \frac{\rho t A_e}{3} I_6$ is often used for explicit dynamics.

6. *Global finite-element system*

After assembling all element matrices:

$$M\ddot{U} + C\dot{U} + KU = F(t) \tag{14}$$

The meaning of each variable is explained on section 2.2.1.6.

2.3 Implicit Newmark-$\boldsymbol{\beta}$ Method for Dynamic Analysis

2.3.1 Newmark time integration formulation

The general form of the dynamic equilibrium equation for the finite element system is shown in equation (9) and (14).

The Newmark-β method assumes the following relationships between displacement, velocity, and acceleration at each time step:

$$U_{n+1} = U_n + \Delta t \dot{U}_n + \frac{\Delta t^2}{2}\left[(1-2\beta)\ddot{U}_n + 2\beta\ddot{U}_{n+1}\right] \tag{15}$$

$$\dot{U}_{n+1} = \dot{U}_n + \Delta t\left[(1-\gamma)\ddot{U}_n + \gamma\ddot{U}_{n+1}\right] \tag{16}$$

where:

- β and γ are method parameters that determine stability and accuracy.
- Δt is the time step size
- n is the current time step, $n+1$ is the next time step

For the Implicit Newmark-β method:

- $\beta = \frac{1}{4}$ and $\gamma = \frac{1}{2}$ (midpoint rule).

2.3.2 Rearranging for acceleration (predictor-corrector)

Solving the acceleration equation for $\ddot{U}_{n+1}$

$$\ddot{U}_{n+1} = \frac{1}{\beta\Delta t^2}\left[U_{n+1} - U_n - \Delta t\dot{U}_n - \frac{\Delta t^2}{2}(1-2\beta)\ddot{U}_n\right] \tag{17}$$

Substituting this into velocity equation

$$\dot{U}_{n+1} = \dot{U}_n + \gamma\Delta t\ddot{U}_{n+1} + (1-\gamma)\Delta t\ddot{U}_n \tag{18}$$

2.3.3 Forming the effective stiffness matrix

Substituting $\ddot{U}_{n+1}$ into the dynamic equilibrium equation gives:

$$M\frac{1}{\beta\Delta t^2}\left(U_{n+1} - U_n - \Delta t\dot{U}_n - \frac{\Delta t^2}{2}(1-2\beta)\ddot{U}_n\right) + C\left(\gamma\frac{U_{n+1}-U_n}{\Delta t} + (1-\gamma)\dot{U}_n\right) + KU_{n+1} = F_{n+1}$$

Rearrange terms:

$$\left[K + \frac{\gamma}{\beta\Delta t}C + \frac{1}{\beta\Delta t^2}M\right]U_{n+1} = F_{n+1} + R_n \tag{19}$$

where R_n is the accumulated effect of the previous time step:

$$R_n = M\left[\frac{1}{\beta\Delta t^2}U_n + \frac{1}{\beta\Delta \mathrm{t}}\dot{U}_n + \frac{1-\beta}{2\beta}\ddot{U}_n\right] + C\left[\frac{\gamma}{\beta\Delta t} + \frac{\gamma}{\beta}\dot{U}_n + \frac{\gamma-1}{\beta}\ddot{U}_n\right] \quad (20)$$

Defining the effective force vector:

$$F_{eff} = F_{n+1} + R_n$$

This is a linear system solved at each time step:

$$K_{eff}U_{n+1} = F_{eff}$$

2.3.4 Stability and accuracy

- For $\beta = 1/4\,, \gamma = 1/2$, the method is unconditionally stable.
- Higher accuracy is achieved by choosing a smaller Δt values, but this increases computational cost.

3 Numerical Implementation

- Language: Python ≥ 3.8
- Libraries:
  - Numpy
  - Pandas
  - Scipy
  - Matplotlib
  - Openpyxl
  - python

- Solver:
  - Static: Global $\boldsymbol{KU} = \boldsymbol{F}$
  - Dynamic: Implicit Newmark-β ($\beta = 1/4,\ \gamma = 1/5$) with lumped and proportional damping
- Code structure: “fem2d” package (mesh → assembly → solver → post-processing)